

CHEM 301 Organic Chemistry I

Problem Set 4

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Assigned: October 2, 2026 Due: October 9, 2026 Returned: October 14, 2026

Score: 47 of 50

1. Rank the following carbocations by stability and explain: methyl, primary, secondary, tertiary.

Answer: Tertiary is most stable, then secondary, primary, and methyl is least stable. Stability increases with more alkyl groups because they donate electron density through hyperconjugation and the inductive effect, spreading out the positive charge.

2. Predict the major product of the addition of HBr to 2-methyl-2-butene and name the rule used.

Answer: The major product is 2-bromo-2-methylbutane. By Markovnikov's rule, the hydrogen adds to the carbon with more hydrogens and the bromine adds to the more substituted carbon, forming the more stable tertiary carbocation intermediate.

3. Identify whether SN1 or SN2 dominates for a tertiary alkyl halide in a polar protic solvent.

Answer: SN1 dominates. Tertiary substrates are too hindered for the backside attack of SN2, and the polar protic solvent stabilizes the carbocation intermediate, favoring the two step SN1 pathway.

4. Draw the Newman projection note: describe the most stable conformation of butane along the C2 to C3 bond.

Answer: The most stable conformation is anti, where the two methyl groups are 180 degrees apart. This minimizes steric strain. The gauche conformation, 60 degrees, is higher in energy, and the fully eclipsed is highest.

5. Explain why benzene does not undergo typical addition reactions like an alkene.

Answer: Benzene is aromatic and its six pi electrons are delocalized in a stable ring. Addition would break that aromatic stabilization, so benzene instead undergoes substitution, which preserves the aromatic system.

Grader note: Question 4 lost 3 points, energy ordering correct but gauche angle mislabeled. Good work overall. R.O.